a portfolio that is changing over time is computed by summing the portfolio variance over each period. The total risk borne by a portfolio manager over a multi-period horizon is thus determined as follows:

Let r_k be the vector of shares held in the portfolio at time k, e.g.,

$$r_k = \begin{pmatrix} r_{1k} \\ r_{2k} \\ \vdots \\ r_{mk} \end{pmatrix} \tag{10.12}$$

where r_{ij} is the number of shares of stock i held in the portfolio at the beginning of period j. That is,

$$r_{ij} = \sum_{k=1}^{j} x_{ik} \tag{10.13}$$

Notice that this is the reverse notation used for residual shares in the trade schedule optimization. Then the total risk borne by the portfolio manager over the entire T-period horizon is:

$$\sigma_p^2 = \underbrace{r_1' C^* r_1 + \cdots + r_n' C^* r_n}_{\textit{Trading Horizon}} + \underbrace{r_{n+1}' C^* r_{n+1} + \cdots + r_t' C^* r_t}_{\textit{Holding Period}} \tag{10.14}$$

where C^* is the covariance matrix expressed in $(\$/\text{shares})^2$ scaled to the length of the trading period. For example, if each trading interval is fifteen minutes and C is the annualized covariance matrix, we have $C^* = \frac{1}{250} \cdot \frac{1}{26} \cdot C$ since there are approximately 250 trading days in a year and 26 fifteen minute intervals in a day. Now, since there are no additional transactions in the portfolio after the end of the trading horizon (e.g., k = n + 1,. . ., t) we have $r_{in+1} = r_{in+2} = \cdots = r_{it} = S_i$ for all stocks. In compressed form, Equation 10.14 is written as:

$$\sigma_p^2 = \underbrace{\sum_{j=1}^{n} r_j' C^* r_j}_{\textit{Trading Horizon}} + \underbrace{(t-n) S' C^* S}_{\textit{Holding Period}} \tag{10.15}$$

since there are n trading periods and $(t-n)$ periods where the portfolio is held and unchanged.

The full investment optimization incorporating market impact and timing risk can now be expressed as follows:

$$\underset{x}{Max} \quad \sum_{i=1}^{m} \left(S_i P_{it} - S_i P_{i0} - \sum_{j=1}^{n} x_{ij} MI(x_{ij}) \right) - \lambda \cdot \left(\sum_{j=1}^{n} r_j' C^* r_j + (t-n) S' C^* S \right) \tag{10.16}$$